

```
#include<stdio.h>
#include<math.h>
#define QSIZE 4
#define TRUE 1
#define FALSE 0

struct Queue
{
int rear, front;
struct student
{
int regno;
char name[15];
int marks;
}st[QSIZE];
};

struct Queue q;

void insert();
void delete();

int empty();

void main()
{
char ch;
int c,x,i;
q.rear=-1;
q.front=0;
do
{
printf("MENU\n");
printf("1. INSERT \n 2. DELETE\n 3. Display\n ");
scanf("%d",&c);
switch(c)
{
```

```

case 1: insert();
        break;
case 2: delete();
        break;
case 3: printf("elements are\n");
        for(i=q.rear;i>=q.front;i--)
        {
            printf("%d ", q.st[i].regno);
            printf("%s ", q.st[i].name);
            printf("%d ", q.st[i].marks);
        }
        break;
}
printf("do you wish to continue\n");
scanf(" %c",&ch);
}while(ch!='n');
}

```

```

void insert()
{
if(q.rear==QSIZE-1)
{
printf("Queue overflow\n");
}
else
{
q.rear++;
printf("enter regno\n");
scanf("%d",&q.st[q.rear].regno);
scanf("%s",q.st[q.rear].name);
scanf("%d",&q.st[q.rear].marks);
}
}

```

```
int delete()
{
if(empty())
{
printf("Queue underflow\n");
}
else
{
printf("deleted elements are\n");
printf("%d",q.st[q.front].regno);
printf("%s",q.st[q.front].name);
printf("%d",&q.st[q.front].marks);
q.front++;
}
}
int empty()
{
if (q.rear<q.front)
return TRUE;
else
return FALSE;
}
```